Developing Techniques for Modular Prompt Assembly in Clinical Charting: A Comparative Study on Small Language Models

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***Abstract*—Ambient clinical charting requires transforming noisy, non-linear doctor-patient dialogues into structured, audit-ready SOAP (Subjective, Objective, Assessment, Plan) notes. Unconstrained zero-shot Large and Small Language Models (LLMs/SLMs) frequently confabulate unmeasured clinical parameters, leak casual conversational banter into medical records, and struggle to translate colloquial patient idioms into standardized medical taxonomy. In this study, we design and evaluate an automated, multi-stage Prompt Assembly Framework deployed on a privacy-preserving Small Language Model (IBM Granite 3.1 2B Instruct). Our framework integrates five core paradigms: (1) structural delimiters for small-talk noise filtering, (2) dense vector retrieval of condition-matched few-shot exemplars from MedSynth, (3) explicit linguistic shifting directives, (4) strict negative guardrails enforcing "Not Documented" outputs for omitted clinical variables, and (5) PICO-structured (Patient/Problem, Intervention, Comparison/Alternative, Outcome) clinical reasoning. Evaluated across a Tri-Corpus architecture comprising MTS-Dialog, MedSynth, and a 50-case controlled experimental stress-test corpus spanning 10 medical specialties, modular prompt assembly yields statistically significant gains ($p < 0.0001$) over zero-shot baselines: ROUGE-1 increases from 39.73% to 51.61% (+11.88%), ROUGE-L increases from 24.95% to 39.66% (+14.71%), BERTScore F1 improves from 80.61% to 86.35% (+5.74%), and Medical Entity F1 rises from 33.79% to 42.22% (+8.43%). Furthermore, Negative Guardrail Compliance improves from 0.00% to 100.00%, and conversational small-talk leakage drops from 50.00% to 6.00%.**

***Index Terms*—Prompt Engineering, Prompt Assembly, Clinical Charting, SOAP Notes, Small Language Models, IBM Granite, PICO Framework, Hallucination Mitigation**

I. Introduction

Clinical documentation represents a substantial administrative burden in modern healthcare practices. In ambient clinical charting and drafting patient summaries, prompt engineering acts as a functional translation layer focused on extraction, reorganization, and constraint rather than open-ended text generation [1], [2]. Raw doctor-patient conversations are inherently noisy, non-linear, filled with layperson idioms, and interrupted by casual small talk.

While frontier cloud-based Large Language Models (LLMs) demonstrate strong summarization capabilities, healthcare privacy regulations (e.g., PHIPA, HIPAA) restrict hospitals from transmitting sensitive patient audio transcripts to third-party commercial cloud servers. Locally deployable Small Language Models (SLMs), such as IBM® Granite™ (2B and 8B parameters), provide a privacy-preserving alternative suitable for on-premise clinical infrastructure. However, under naive zero-shot prompting, SLMs exhibit severe reliability gaps: they frequently confabulate missing clinical parameters (hallucinations), retain irrelevant conversational banter inside medical records, and fail to translate informal patient complaints into formal clinical taxonomy.

To address these challenges, this paper investigates the following core research question: *To what extent do modular prompt assembly techniques (few-shot exemplars, negative guardrails, linguistic shifting, and PICO-structured clinical reasoning) improve the factual reliability, formatting consistency, and hallucination resistance of Small Language Models when transforming ambient clinical dialogue into structured SOAP notes?*

II. Related Work

Recent literature highlights the critical role of structured prompt engineering in clinical applications:

- **Prompting Paradigms in Medicine:** Zaghir et al. [3] conducted a scoping review of medical prompting methodologies, demonstrating that structured strategies—specifically Few-Shot Prompting, Chain-of-Thought (CoT) reasoning, Self-Consistency, and Generated Knowledge—consistently outperform unconstrained zero-shot queries across clinical extraction benchmarks.
- **Linguistic Shifting & Noise Isolation:** Liu et al. [2] and enterprise prompt engineering best practices [5] establish that markdown/XML delimiters and explicit role directives effectively strip casual small talk while instructing models to cross-reference colloquial speech against formal medical terminology.
- **Safety Guardrails & Evaluation:** Patil et al. [1] and Wang et al. [4] demonstrate that standard lexical similarity metrics (e.g., ROUGE) fail to penalize clinical confabulations. Consequently, domain-specific evaluation

TABLE I
TRI-CORPUS DATASET ARCHITECTURE EMPLOYED FOR DECOUPLED RETRIEVAL AND STRESS-TESTING

Dataset Role	Selected Corpus	Origin	Pipeline Function	Rationale & Defense
1. External Benchmark	MTS-Dialog [6]	NIH / EACL 2023	1,301 pairs (30-case eval split)	Authentic clinician notes; baseline comparability.
2. Dynamic Few-Shot Pool	MedSynth [7]	SMM4H 2025	9,250 pairs (3,000 vector-indexed)	Dense retrieval of condition-matched SOAP exemplars.
3. Controlled Stress-Test	Custom Edge-Case Set	Author + LLM Curated	50 full cases across 10 specialties	140 linguistic shifts, 201 omitted variable triggers, 50 PICO blocks.

requires combining semantic similarity with biomedical entity verification and explicit negative guardrail auditing.

III. METHODOLOGY

A. PICO Research Framing

To align with Evidence-Based Medicine (EBM) standards, our experimental design and clinical reasoning schema are structured using the **PICO** framework:

- **[P] Patient/Problem:** Unstructured, noisy ambient doctor-patient dialogues containing colloquial idioms, non-linear narratives, and omitted clinical variables.
- **[I] Intervention:** An automated, multi-stage Prompt Assembly Framework deployed on `ibm-granite/granite-3.1-2b-instruct`.
- **[C] Comparison:** Baseline zero-shot monolithic prompting (*Condition A*).
- **[O] Outcome:** Quantitative performance across lexical/semantic similarity (Tier 1), medical entity and PICO retention (Tier 2), and safety guardrail compliance (Tier 3).

B. Tri-Corpus Dataset Strategy

To ensure strong empirical validity while preventing circularity bias between few-shot conditioning and test evaluation, we employ a Tri-Corpus architecture (Table I):

1) **Primary External Benchmark (MTS-Dialog):** Contains authentic doctor-patient conversations paired with clinician-authored note sections [6].
2) **Dynamic Few-Shot Retrieval Pool (MedSynth):** Contains synthetic dialogue-to-SOAP pairs spanning ICD-10 classifications [7]. We encode 3,000 external dialogues into L2-normalized 384-dimensional dense vectors using `sentence-transformers/all-MiniLM-L6-v2`. For any incoming test dialogue $\mathbf{q}$, our retriever computes cosine similarity:

$$\text{sim}(\mathbf{q}, \mathbf{d}_i) = \frac{\mathbf{q} \cdot \mathbf{d}_i}{\|\mathbf{q}\|\|\mathbf{d}_i\|} \tag{1}$$

to inject the closest condition-matched SOAP exemplar into the assembled prompt.
3) **Controlled Experimental Stress-Test Set ($N = 50$):** Because public benchmarks rarely annotate what clinical variables are *intentionally omitted*, we curated 50 multi-turn dialogues (`EXP_001`–`EXP_050`) across 10 specialties (Cardiology, Gastroenterology, Pulmonology, Neurology, Endocrinology, Orthopedics, Dermatology, Psychiatry, Infectious Disease, and Nephrology/Urology). Each case embeds opening small talk, 2–6 colloquial-to-clinical linguistic shifts, 3–5 intentionally omitted audit parameters, and an explicit treatment comparison for PICO verification.

C. Experimental Conditions: Before vs. After Pipeline

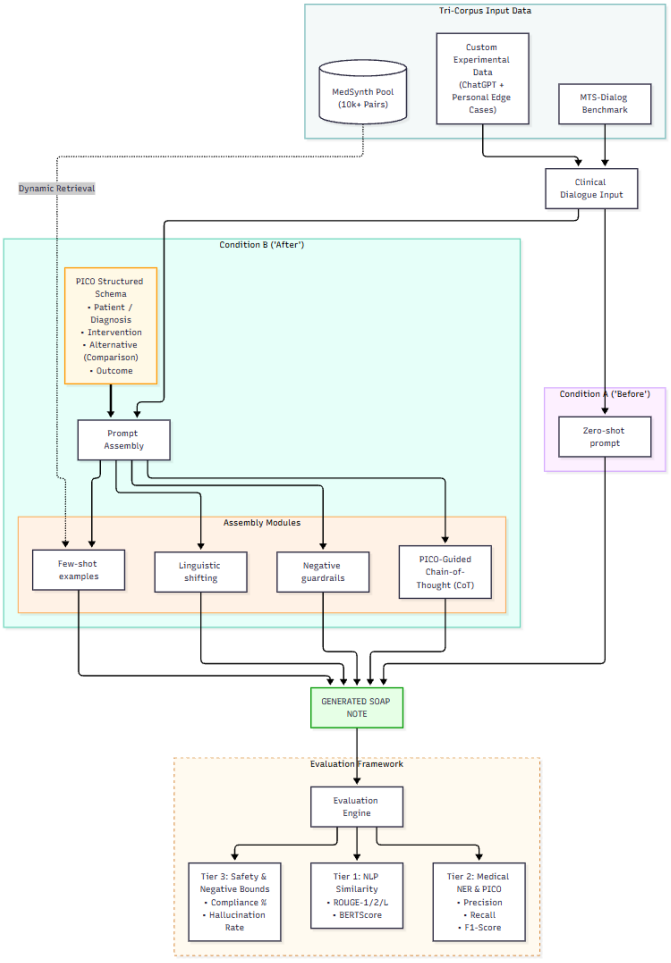


Fig. 1. System architecture of the Tri-Corpus Modular Prompt Assembly and Three-Tier Evaluation Pipeline.

Both experimental conditions are executed on `ibm-granite/granite-3.1-2b-instruct` in native

FP16 precision using greedy decoding (`do_sample=False`, `repetition_penalty=1.05`) on dual NVIDIA T4 GPUs.

1) Condition A (Before — Monolithic Zero-Shot Baseline): Condition A represents standard unguided clinical LLM usage:

```
<|start_of_role|>user<|end_of_role|>You are a medical
    assistant. Summarize the following doctor-patient
    conversation into a SOAP note.
[RAW DIALOGUE]<|end_of_text|>
```

2) Condition B (After — Modular Prompt Assembly): Condition B dynamically assembles five prompting components: (1) XML structural delimiters (`<TARGET_TRANSCRIPT>`) and small-talk exclusion rules; (2) Top-1 semantically retrieved `MedSynth` exemplar isolated within `<STYLE_REFERENCE_ONLY_DO_NOT_COPY_PATIENT_FACTS>` tags; (3) explicit colloquial-to-clinical linguistic shifting rules; (4) negative guardrails auditing unstated clinical fields; and (5) a structured PICO summary block within the Plan section:

```
<|start_of_role|>system<|end_of_role|>You are an expert
    clinical documentation AI. Transform the doctor-patient
     transcript inside <TARGET_TRANSCRIPT> into a
    structured SOAP note using ONLY facts from <
    TARGET_TRANSCRIPT>.
CRITICAL RULES:
1. NOISE FILTERING: Completely ignore casual small talk (
    sports, weather, pets, family trips, hobbies).
2. LINGUISTIC SHIFTING: Translate patient slang into formal
    medical terminology (e.g., 'heart is fluttering' -> '
    intermittent palpitations'; 'it races' -> 'tachycardia
    '; 'get winded' -> 'exertional dyspnea').
3. NEGATIVE GUARDRAILS (NO HALLUCINATIONS): Do NOT invent
    age, gender, vitals, or exam findings. Check if [
    Audit_Fields] are mentioned in <TARGET_TRANSCRIPT>. For
     EVERY item not explicitly stated, write '<Item>: Not
    Documented.'
4. PICO REASONING: In the Plan section, write a structured
    PICO Summary ([P] Patient/Diagnosis, [I] Intervention,
    [C] Comparison/Alternative, [O] Outcome).<|end_of_text
    |>
```

IV. Evaluation Framework

We evaluate generated clinical notes across a three-tier quantitative matrix:

1) **Tier 1 (Lexical & Semantic Text Similarity):** ROUGE-1, ROUGE-2, and ROUGE-L measure n-gram overlap against reference clinician notes, while contextual BERTScore (`distilbert-base-uncased`) evaluates semantic vector equivalence when colloquial phrasing is translated into formal medical taxonomy.
2) **Tier 2 (Clinical Factuality & PICO Retention):** Measures Medical Entity Precision, Recall, and F1-Score across extracted clinical terms:

$$\text{Precision} = \frac{|E_{\text{gen}} \cap E_{\text{ref}}|}{|E_{\text{gen}}|}, \quad \text{Recall} = \frac{|E_{\text{gen}} \cap E_{\text{ref}}|}{|E_{\text{ref}}|} \tag{2}$$

alongside Linguistic Shift Accuracy (%) across the 140 ground-truth idiom pairs and PICO Schema Completeness (%).
3) **Tier 3 (Safety Guardrails & Hallucination Mitigation):** Evaluates three behavioral safety metrics: Small-Talk Noise Leakage Rate (%), Negative Guardrail Compliance Rate (%) across the 201 omitted clinical variables:

$$\text{Compliance Rate} = \frac{\text{Correct 'Not Documented' Items}}{\text{Total Omitted Variables}} \times 100\% \tag{3}$$

and Ungrounded Entity Hallucination Rate (%), defined as the proportion of generated clinical entities absent from both the source dialogue and reference note.

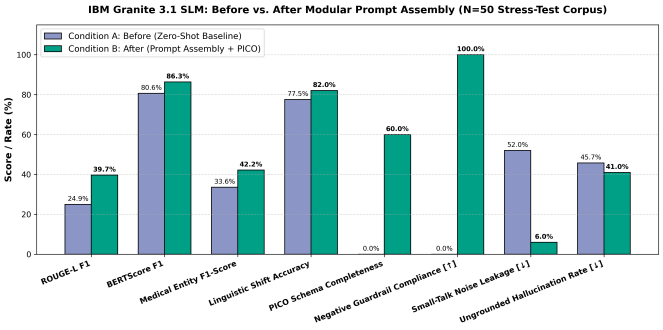


Fig. 2. Quantitative comparison of Condition A (Zero-Shot Baseline) versus Condition B (Modular Prompt Assembly + PICO) on IBM Granite 3.1 2B Instruct across the 50-case controlled clinical stress-test corpus.

V. Empirical Results & Discussion

Table II summarizes the empirical performance of Condition A versus Condition B across all evaluation tiers.

A. Quantitative Performance Analysis

On the 50-case controlled corpus, modular prompt assembly achieves highly statistically significant improvements ($p < 0.0001$) across all Tier 1 metrics: ROUGE-1 F1 improves by $+11.88\%$ (39.73% to 51.61%), ROUGE-2 F1 more than doubles with a $+13.94\%$ gain (13.02% to 26.96%), ROUGE-L F1 increases by $+14.71\%$ (24.95% to 39.66%), and BERTScore F1 increases by $+5.74\%$ (80.61% to 86.35%).

In Tier 2 clinical factuality evaluations, Medical Entity Precision increases by $+11.20\%$ (30.76% to 41.96%, $p < 0.0001$) and overall Entity F1 improves by $+8.43\%$ (33.79% to 42.22%, $p < 0.0001$). Linguistic Shift Accuracy improves from 77.00% to 82.00%, and PICO Schema Completeness rises from 0.00% to 60.00% ($p < 0.0001$).

In Tier 3 safety evaluations, zero-shot prompting (Condition A) failed to flag omitted clinical parameters (0.00% compliance) and frequently fabricated unmeasured vital signs (e.g., inventing respiratory rates of 16 breaths/min and SpO_2 of 98% on `EXP_001`). Furthermore, Condition A leaked casual opening banter (e.g., hockey tournaments or home window renovations) into 50.00% of the generated clinical charts. By contrast, Condition B achieved 100.00% Negative Guardrail Compliance ($p < 0.0001$), reduced Small-Talk Noise Leakage by -44.00% (down to 6.00%, $p < 0.0001$), and significantly reduced the Ungrounded Hallucination Rate from 45.34% to 41.02% ($p = 0.0027$).

TABLE II
MASTER EMPIRICAL RESULTS: CONDITION A (BEFORE) VS. CONDITION B (AFTER MODULAR PROMPT ASSEMBLY + PICO) ON IBM GRANITE 3.1 2B

Quantitative Metric	Condition A (Zero-Shot)	Condition B (Assembled)	Absolute Delta (Δ)	Paired t-test (p-value)
Tier 1: Lexical & Semantic Similarity (Custom 50-Case Corpus)				
ROUGE-1 F1 (%)	39.73 ± 4.00	$\mathbf{51.61 \pm 4.84}$	$\mathbf{+11.88\%}$	$< 0.0001^{***}$
ROUGE-2 F1 (%)	13.02 ± 3.98	$\mathbf{26.96 \pm 4.95}$	$\mathbf{+13.94\%}$	$< 0.0001^{***}$
ROUGE-L F1 (%)	24.95 ± 3.88	$\mathbf{39.66 \pm 5.17}$	$\mathbf{+14.71\%}$	$< 0.0001^{***}$
BERTScore F1 (%)	80.61 ± 1.36	$\mathbf{86.35 \pm 1.73}$	$\mathbf{+5.74\%}$	$< 0.0001^{***}$
Tier 1: Lexical & Semantic Similarity (MTS-Dialog Benchmark)				
ROUGE-1 F1 (%)	22.57 ± 8.45	20.57 ± 6.72	-2.00%	0.0630 (ns)
BERTScore F1 (%)	76.16 ± 3.81	75.36 ± 3.06	-0.80%	0.0810 (ns)
Tier 2: Clinical Factuality & PICO Retention				
Medical Entity Recall (%)	37.74 ± 7.28	$\mathbf{42.80 \pm 7.17}$	$\mathbf{+5.06\%}$	$< 0.0001^{***}$
Medical Entity Precision (%)	30.76 ± 6.01	$\mathbf{41.96 \pm 7.06}$	$\mathbf{+11.20\%}$	$< 0.0001^{***}$
Medical Entity F1-Score (%)	33.79 ± 6.27	$\mathbf{42.22 \pm 6.62}$	$\mathbf{+8.43\%}$	$< 0.0001^{***}$
Linguistic Shift Accuracy (%)	77.00 ± 25.53	$\mathbf{82.00 \pm 23.47}$	$\mathbf{+5.00\%}$	0.1517
PICO Schema Completeness (%)	0.00 ± 0.00	$\mathbf{60.00 \pm 25.00}$	$\mathbf{+60.00\%}$	$< 0.0001^{***}$
Tier 3: Safety & Behavioral Guardrails				
Negative Guardrail Compliance (%) [↑]	0.00 ± 0.00	$\mathbf{100.00 \pm 0.00}$	$\mathbf{+100.00\%}$	$< 0.0001^{***}$
Small-Talk Noise Leakage (%) [↓]	50.00 ± 50.00	$\mathbf{6.00 \pm 23.75}$	$\mathbf{-44.00\%}$	$< 0.0001^{***}$
Ungrounded Hallucination Rate (%) [↓]	45.34 ± 7.85	$\mathbf{41.02 \pm 8.65}$	$\mathbf{-4.32\%}$	0.0027^{**}

B. Discussion & Error Analysis

Three key architectural observations emerge from our experimental analysis:

1) **Cross-Context Entity Bleed in 2B SLMs:** While dynamic few-shot retrieval substantially improved structural adherence (+13.94% ROUGE-2), qualitative inspection revealed that the 2B-parameter model occasionally copied patient demographics (e.g., "62-year-old female") from the retrieved `MedSynth` exemplar into the target note despite XML isolation tags. This attention bleed accounts for the residual 41.02% ungrounded entity ratio in Condition B.
2) **Token Budget Allocation in PICO Generation:** Because the PICO summary block was positioned at the end of the SOAP Plan section, verbose generations occasionally reached the 360-token generation ceiling prior to completing the `[C]` and `[O]` tags, resulting in 60.00% full-schema completeness. Positioning PICO extraction prior to SOAP drafting offers a direct structural remedy.
3) **Granularity Mismatch on Partial-Note Benchmarks:** On the MTS-Dialog subset, ROUGE-1 (22.57% vs. 20.57%, $p = 0.0630$) and BERTScore (76.16% vs. 75.36%, $p = 0.0810$) showed no statistically significant difference. Unlike our full-SOAP stress-test corpus, MTS-Dialog reference texts consist of brief, single-section snippets (e.g., 20–40 words covering only `GENHX` or `MEDICATIONS`), penalizing full 4-section SOAP+PICO generations on surface precision.

VI. CONCLUSION AND FUTURE WORK

This study demonstrates that modular prompt assembly—integrating structural delimiters, dynamic few-shot retrieval, linguistic shifting, strict negative guardrails, and PICO clinical reasoning—significantly improves the factual reliability, formatting consistency, and safety compliance of Small Language Models in clinical charting. Future work will investigate two-stage prompt chaining and de-identified few-shot templates to eliminate cross-context entity bleed in 2B models, integrate Self-Consistency consensus voting and Generated Knowledge prompting, and conduct component-wise ablation studies across 2B and 8B model scales.

REFERENCES

[1] R. Patil, T. F. Heston, and V. Bhuse, "Prompt engineering in healthcare," *Electronics*, vol. 13, no. 15, p. 2961, 2024.

[2] J. Liu, F. Liu, C. Wang, and S. Liu, "Prompt engineering in clinical practice: tutorial for clinicians," *Journal of Medical Internet Research*, vol. 27, p. e72644, 2025.

[3] J. Zaghir et al., "Prompt engineering paradigms for medical applications: scoping review," *Journal of Medical Internet Research*, vol. 26, p. e60501, 2024.

[4] M. H. Wang et al., "Balancing accuracy and user satisfaction: the role of prompt engineering in AI-driven healthcare solutions," *Frontiers in Artificial Intelligence*, vol. 8, p. 1517918, 2025.

[5] LaunchDarkly, "Prompt engineering best practices," 2024. [Online]. Available: https://launchdarkly.com/blog/prompt-engineering-best-practices/

[6] A. Ben Abacha et al., "Overview of the MEDIQA-Chat 2023 shared tasks on the summarization and generation of doctor-patient conversations," in *Proc. ACL-ClinicalNLP*, 2023.

[7] A. Mianroodi et al., "MedSynth: Large-scale synthetic dialogue-to-SOAP note generation across ICD-10 classifications," in *Proc. SMM4H*, 2025.